

Esp32Tap — Depth-First Cluster Build and Test

Guide mode: Empty-board build · Build one cluster, prove it, sign it, then unlock the next.

No treadmill cable before clusters 1–10 pass; do not connect a treadmill cable until Cluster 11. No exact physical placement is prescribed. Follow the named parts, pins, nets, and colors.

Device serial:	Operator:
Hardware revision:	Date:

Source manifest: esp32tap-breadboard-netlist.csv; esp32tap-breadboard-clusters.html; esp32tap-breadboard-from-to.html; DigiKey invoice 129832848; superseded module checklist used only for already-proven limits and sequences.

Source contract

Mutually exclusive source rule: USB and STANDALONE POWER are mutually exclusive—one source only. USB connected means STANDALONE POWER physically removed. STANDALONE POWER installed means USB physically disconnected. All sources off before any RJ45 attachment/removal, harness change, or direct-path change. COIL POWER stays open through the unloaded test.

Safety contract

- Initial source: 8.00 V, initial 250 mA current limit; coil-open current below 50 mA. VIN must be 7.20–7.90 V and LOGIC_3V3 must be 3.20–3.40 V.
- TREAD_OK is low below UV, high at 8.00 V, and low above OV. UV rising is 6.25–6.55 V; OV falling is 10.30–10.90 V.
- TPS709 enabled: 4.75–5.25 V; TPS709 disabled: below 0.25 V. Loaded relay current limit no more than 500 mA.
- Coil: 90–110 mA and at least 4.50 V. BC337 VCE no more than 0.30 V. Feedback (1,0) energized and (0,1) bypass; 00 or 11 is a fault.
- Removal of USB logic power, GPIO21 command, TREAD_OK, or VIN must restore NC bypass within 100 ms.
- Five-minute hold: TPS709 and BC337 no more than 45°C and no more than 10°C over ambient.
- Any unexpected reset is a STOP. No treadmill relay transfer or MCU TX is permitted by this guide.

Firmware: The current no-control diagnostic cannot pass a relay exercise gate and cannot pass a UART exercise gate. Relay/coil/TX work requires a bounded relay exerciser with exact identity. Standalone/bypass requires an observer identity and manifest proving relay and TX disabled.

Wire palette: BLACK GND · RED +8V_RAW/FUSED_8V · ORANGE VIN · BLUE TSR_3V3/LOGIC_3V3 · VIOLET +5V_RLY/coil · YELLOW control/UART · GREEN feedback/RX · WHITE direct pass-through.

Cluster 1 — Raw protection

Dependencies: Empty board and verified bench equipment
Inputs: +8V_RAW, GND

Outputs: VIN, FUSED_8V
Source state: USB absent; STANDALONE POWER removed;
COIL POWER removed; bench 8.00 V, initial 250 mA limit

Parts

F1 RXEF075 · 750 mA hold · RXEF075HF-ND · non-polar. **D1** 1N5822-TP Schottky · 1N5822-TPMSCT-ND · cathode stripe faces VIN. **D2** P6KE12A-TP · P6KE12A-TPMSCT-ND · cathode to VIN. **CIN** 22 μ F 25 V · 732-8821-1-ND · negative stripe to GND.

Operator part record: F1 · RXEF075 · 750 mA hold · RXEF075HF-ND · non-polar

Operator wiring record: F1 · lead 1 · +8V_RAW · RED; F1 · lead 2 · FUSED_8V · RED; D1 · anode · FUSED_8V · RED; D1 · cathode · VIN · ORANGE; D2 · cathode · VIN · ORANGE; D2 · anode · GND · BLACK; CIN · positive · VIN · ORANGE; CIN · negative · GND · BLACK

Build — exact pin/net/color wiring

Step	Ref	Color	From	To	Part	Note
1	15	RED	+8V_RAW	RXEF075 lead 1	Wire	Local electronics tap only Connect F1 lead 1 to +8V_RAW using RED.
2	16	RED	RXEF075 lead 2	1N5822 anode	Wire	Non-striped diode end Connect F1 lead 2 to FUSED_8V using RED. Connect D1 anode to FUSED_8V using RED.
3	17	ORANGE	1N5822 cathode	VIN	Wire	Striped diode end Connect D1 cathode to VIN using ORANGE.
4	18	ORANGE	VIN	P6KE12A cathode	Wire	Striped TVS end Connect D2 cathode to VIN using ORANGE.
5	19	BLACK	P6KE12A anode	GND	Wire	TVS return Connect D2 anode to GND using BLACK.
6	23	ORANGE	VIN	22 μ F positive	Wire	Bulk capacitor Connect CIN positive to VIN using ORANGE.
7	24	BLACK	22 μ F negative	GND	Wire	Observe polarity Connect CIN negative to GND using BLACK.

Unpowered test

Instruction / kind / points / expected	Evidence
Check polarity and resistance · polarity · D1 cathode/VIN, D2 cathode/VIN, CIN negative/GND, VIN–GND · stripes correct and no short	

Powered test

Instruction / input / output / evidence / limits	Measured
Apply protected raw power · +8V_RAW at 8.00 V · FUSED_8V and VIN · record volts/current · VIN 7.20–7.90 V; coil-open current below 50 mA	

STOP gate: Current approaches 250 mA, VIN is outside 7.20–7.90 V, polarity is wrong, or anything warms. Remove power; do not build Cluster 2.

PASS gate: Protection chain, polarity, VIN window, and coil-open current pass. **PASS — continue** to Cluster 2 only.

Operator:

Date:

Signed PASS:

Cluster 2 — TSR supply

Dependencies: Cluster 1
Inputs: VIN, GND

Outputs: TSR_3V3
Source state: Cluster 1 source state retained; STANDALONE POWER remains removed

Parts

U2 TSR 1-2433E · 3.3 V converter · 1951-TSR1-2433E-ND · pin order verified. **C2/C3** 10 μ F 50 V electrolytics · 1189-2322-ND · negative stripes to GND. **JP1** removable STANDALONE POWER · NOT ORDERED · leave removed.

Operator part record: U2/C2/C3/JP1 · TSR 1-2433E; two electrolytics; STANDALONE POWER · 3.3 V; 10 μ F 50 V; removable link · 1951-TSR1-2433E-ND; 1189-2322-ND; NOT ORDERED · U2 pin order verified; capacitor negatives to GND; link removed

Operator wiring record: U2 · pin 1 · VIN · ORANGE; C2 · positive · VIN · ORANGE; U2 · pin 2 · GND · BLACK; C2 · negative · GND · BLACK; U2 · pin 3 · TSR_3V3 · BLUE; C3 · positive · TSR_3V3 · BLUE; C3 · negative · GND · BLACK; JP1 · between · TSR_3V3 and LOGIC_3V3 · BLUE

Build — exact pin/net/color wiring

Step	Ref	Color	From	To	Part	Note
1	20	ORANGE	VIN	TSR1-2433E pin 1	Wire	Converter input Connect U2 pin 1 to VIN using ORANGE.
2	25	ORANGE	VIN	10 μ F input-cap positive	Wire	Near TSR input Connect C2 positive to VIN using ORANGE.
3	21	BLACK	TSR1-2433E pin 2	GND	Wire	Converter ground Connect U2 pin 2 to GND using BLACK.
4	26	BLACK	10 μ F input-cap negative	GND	Wire	Observe polarity Connect C2 negative to GND using BLACK.
5	22	BLUE	TSR1-2433E pin 3	TSR_3V3	Wire	Converter output Connect U2 pin 3 to TSR_3V3 using BLUE.
6	27	BLUE	TSR_3V3	10 μ F output-cap positive	Wire	Near TSR output Connect C3 positive to TSR_3V3 using BLUE.
7	28	BLACK	10 μ F output-cap negative	GND	Wire	Observe polarity Connect C3 negative to GND using BLACK.
8	31	BLUE	TSR_3V3	LOGIC_3V3	STANDALONE POWER jumper	Leave removed when USB is attached Fit JP1 between TSR_3V3 and LOGIC_3V3 using BLUE; leave JP1 removed.

Unpowered test

Instruction / kind / points / expected	Evidence
Verify converter map and capacitor polarity · continuity · U2 pins 1/2/3 to VIN/GND/TSR_3V3 · exact mapping; TSR_3V3 not joined to future LOGIC_3V3	

Powered test

Instruction / input / output / evidence / limits	Measured
Energize verified VIN · VIN · TSR_3V3 · record input/output/current · TSR_3V3 3.20–3.40 V and future LOGIC_3V3 unpowered	

STOP gate: TSR_3V3 is outside 3.20–3.40 V, polarity is wrong, LOGIC_3V3 is energized across the removed link, or current/temperature rises.

PASS gate: Verified Cluster 1 VIN produces isolated TSR_3V3 in range. **PASS — continue** to Cluster 3 only.

Operator: _____ Date: _____ Signed PASS: _____

Cluster 3 — DevKit and logic supply

Dependencies: Cluster 2; Verified USB bench source
Inputs: TSR_3V3, USB_5V, GND

Outputs: LOGIC_3V3, DevKit GND, GPIO21_CMD_POST, GPIO15_TX_ENABLE_POST, GPIO17_TX, GPIO38, USB_5V
Source state: First USB-only with STANDALONE POWER removed; then USB physically disconnected before standalone link installation

Parts

U1 ESP32-S3-DEVKITC-1-N8R8 · 1965-ESP32-S3-DEVKITC-1-N8R8-ND · antenna clear, USB ports identified. **JP1** STANDALONE POWER · NOT ORDERED · removable and labeled. **C4** 100 nF · BC1084CT-ND · non-polar.

Operator part record: U1/JP1/C4 · ESP32-S3-DEVKITC-1-N8R8; STANDALONE POWER; capacitor · DevKit; removable link; 100 nF · 1965-ESP32-S3-DEVKITC-1-N8R8-ND; NOT ORDERED; BC1084CT-ND · antenna clear; USB ports identified; link removable; capacitor non-polar

Operator wiring record: DevKit · 3V3 · LOGIC_3V3 · BLUE; DevKit · GND · GND · BLACK; C4 · lead 1 · LOGIC_3V3 · BLUE; C4 · lead 2 · GND · BLACK; TSR_3V3 · endpoint · the JP1 standalone-side post · BLUE; LOGIC_3V3 · endpoint · the JP1 logic-side post · BLUE; DevKit · GPIO15 · the DevKit-side GPIO15 jumper post · YELLOW; DevKit · GPIO21 · the DevKit-side GPIO21 jumper post · YELLOW

Build — exact pin/net/color wiring

Step	Ref	Color	From	To	Part	Note
1	29	BLUE	DevKit 3V3	LOGIC_3V3	Jumper wire	Logic rail Connect DevKit 3V3 to LOGIC_3V3 using BLUE.
2	30	BLACK	DevKit GND	GND	Jumper wire	Common ground Connect DevKit GND to GND using BLACK.
3	NEW C3-3	BLUE	C4 lead 1	LOGIC_3V3	Wire	Connect C4 lead 1 to LOGIC_3V3 using BLUE.
4	NEW C3-4	BLACK	C4 lead 2	GND	Wire	Connect C4 lead 2 to GND using BLACK.
5	NEW C3-5	BLUE	TSR_3V3 endpoint	the JP1 standalone-side post	Wire	Connect TSR_3V3 endpoint to the JP1 standalone-side post using BLUE.
6	NEW C3-6	BLUE	LOGIC_3V3 endpoint	the JP1 logic-side post	Wire	Connect LOGIC_3V3 endpoint to the JP1 logic-side post using BLUE; leave JP1 removed.
7	NEW C3-7	YELLOW	DevKit GPIO15	the DevKit-side GPIO15 jumper post	Wire	Connect DevKit GPIO15 to the DevKit-side GPIO15 jumper post using YELLOW; label the post.
8	NEW C3-8	YELLOW	DevKit GPIO21	the DevKit-side GPIO21 jumper post	Wire	Connect DevKit GPIO21 to the DevKit-side GPIO21 jumper post using YELLOW; label the post.

Unpowered test

Instruction / kind / points / expected	Evidence
Map source handoff · mapping · DevKit 3V3, LOGIC_3V3, TSR_3V3, JP1 · USB path mapped; JP1 removable; no source backfeed	

Powered test

Instruction / input / output / evidence / limits	Measured
Prove each exclusive source · USB_5V then TSR_3V3 · LOGIC_3V3 · record both runs · LOGIC_3V3 3.20–3.40 V; never energize USB and JP1 together	

STOP gate: Sources overlap, either source backfeeds the other, LOGIC_3V3 is outside 3.20–3.40 V, or DevKit boot is unstable.

PASS gate: USB-only and standalone-only handoff both pass, with the other source physically absent. **PASS — continue** to Cluster 4 only.

Operator:

Date:

Signed PASS:

Cluster 4 — TPS3700 voltage monitor

Dependencies: Cluster 1, Cluster 3
Inputs: VIN, LOGIC_3V3, GND

Outputs: UV_SENSE, OV_SENSE, TREAD_OK, TREAD_OK_MCU
Source state: USB logic only; STANDALONE POWER removed; bench source swept at protected VIN

Parts

U3 TPS3700DDCR · 296-30395-1-ND; **adapter** LCQT-SOT23-6 · A879AR-ND; **header** cut from PRPC040SAAN-RC · S1011EC-40-ND; **RUV** 150 kΩ · 150KXBK-ND; **ROV** 255 kΩ · 13-MFR-25FBF52-255K-ND; 10 kΩ/100 kΩ/4.7 kΩ · NOT ORDERED; two 1 nF · 445-173473-1-ND; 100 nF · BC1084CT-ND.

Adapter gate: continuity-map adapter pins 1–6 to the header before installation. Record the mapped header positions: 1 ____ 2 ____ 3 ____ 4 ____ 5 ____ 6 ____.

Divider formulas: $UV_SENSE = VIN \times 10\text{ k}\Omega / (150\text{ k}\Omega + 10\text{ k}\Omega) = VIN / 16$. $OV_SENSE = VIN \times 10\text{ k}\Omega / (255\text{ k}\Omega + 10\text{ k}\Omega) = VIN / 26.5$.

Operator part record: U3/dividers/filters · TPS3700DDCR plus adapter and UV/OV networks · 150 kΩ/10 kΩ; 255 kΩ/10 kΩ; 1 nF; 100 nF · 296-30395-1-ND; A879AR-ND; 150KXBK-ND; 13-MFR-25FBF52-255K-ND; 445-173473-1-ND; BC1084CT-ND; NOT ORDERED · meter-map adapter pin 1; capacitors non-polar

Operator wiring record: See numbered point-to-point steps below.

Operator header record: HDR3 · PRPC040SAAN-RC header · cut header · S1011EC-40-ND · map before installation

Build — exact pin/net/color wiring

Step	Ref	Color	From	To	Part	Note
1	32	ORANGE	VIN	TPS3700 pin 5 VDD	Wire	Supply Connect U3 pin 5 to VIN using ORANGE; record initials ____.
2	33	BLACK	TPS3700 pin 2	GND	Wire	Ground Connect U3 pin 2 to GND using BLACK; record initials ____.
3	36	WHITE	UV_SENSE	TPS3700 pin 3 INA+	Wire	UV input Connect U3 pin 3 to UV_SENSE using WHITE; record initials ____.
4	43	WHITE	OV_SENSE	TPS3700 pin 4 INB-	Wire	OV input Connect U3 pin 4 to OV_SENSE using WHITE; record initials ____.
5	48	YELLOW	TPS3700 pin 1 OUTA	TREAD_OK	Wire	Open-drain output Connect U3 pin 1 to TREAD_OK using YELLOW; record initials ____.
6	49	YELLOW	TPS3700 pin 6 OUTB	TREAD_OK	Wire	Tie outputs together Connect U3 pin 6 to TREAD_OK using YELLOW; record initials ____.
7	34	ORANGE	VIN	R_UV_TOP 150 kΩ, end 1	Wire	UV divider Connect R_UV_TOP 1 to VIN using ORANGE; record initials ____.
8	35	WHITE	R_UV_TOP end 2	UV_SENSE	Wire	Divider midpoint Connect R_UV_TOP 2 to UV_SENSE using WHITE; record initials ____.
9	37	WHITE	UV_SENSE	R_UV_BOTTOM 10 kΩ, end 1	Wire	Divider bottom Connect R_UV_BOTTOM 1 to UV_SENSE using WHITE; record initials ____.
10	38	BLACK	R_UV_BOTTOM end 2	GND	Wire	Divider return Connect R_UV_BOTTOM 2 to GND using BLACK; record initials ____.
11	39	WHITE	UV_SENSE	C_UV 1 nF, end 1	Wire	Filter Connect C_UV 1 to UV_SENSE using WHITE; record initials ____.
12	40	BLACK	C_UV end 2	GND	Wire	Filter return Connect C_UV 2 to GND using BLACK; record initials ____.
13	41	ORANGE	VIN	R_OV_TOP 255 kΩ, end 1	Wire	OV divider Connect R_OV_TOP 1 to VIN using ORANGE; record initials ____.
14	42	WHITE	R_OV_TOP end 2	OV_SENSE	Wire	Divider midpoint Connect R_OV_TOP 2 to OV_SENSE using WHITE; record initials ____.
15	44	WHITE	OV_SENSE	R_OV_BOTTOM 10 kΩ, end 1	Wire	Divider bottom Connect R_OV_BOTTOM 1 to OV_SENSE using WHITE; record initials ____.
16	45	BLACK	R_OV_BOTTOM end 2	GND	Wire	Divider return Connect R_OV_BOTTOM 2 to GND using BLACK; record initials ____.
17	46	WHITE	OV_SENSE	C_OV 1 nF, end 1	Wire	Filter Connect C_OV 1 to OV_SENSE using WHITE; record initials ____.
18	47	BLACK	C_OV end 2	GND	Wire	Filter return Connect C_OV 2 to GND using BLACK; record initials ____.
19	50	BLUE	LOGIC_3V3	R_TREAD_PULLUP 10 kΩ, end 1	Wire	Pull-up Connect R_TREAD_PULLUP 1 to LOGIC_3V3 using BLUE; record initials ____.
20	51	YELLOW	R_TREAD_PULLUP end 2	TREAD_OK	Wire	Pull-up node Connect R_TREAD_PULLUP 2 to TREAD_OK using YELLOW; record initials ____.
21	52	YELLOW	TREAD_OK	R_TREAD_PULLDOWN 100 kΩ, end 1	Wire	Power-off pull-down Connect R_TREAD_PULLDOWN 1 to TREAD_OK using YELLOW; record initials ____.
22	53	BLACK	R_TREAD_PULLDOWN end 2	GND	Wire	Pull-down return Connect R_TREAD_PULLDOWN 2 to GND using BLACK; record initials ____.
23	54	YELLOW	TREAD_OK	R_TREAD_MCU 4.7 kΩ, end 1	Wire	GPIO isolation Connect R_TREAD_MCU 1 to TREAD_OK using YELLOW; record initials ____.
24	55	YELLOW	R_TREAD_MCU end 2	DevKit GPIO6	Wire	TREAD_OK_MCU Connect R_TREAD_MCU 2 to TREAD_OK_MCU using YELLOW; record initials ____. Connect DEVKIT GPIO6 to TREAD_OK_MCU using YELLOW; record initials ____.
25	56	ORANGE	VIN	C_TPS3700 100 nF, end 1	Wire	Decoupling Connect C_TPS3700 1 to VIN using ORANGE; record initials ____.
26	57	BLACK	C_TPS3700 end 2	GND	Wire	Decoupling return Connect C_TPS3700 2 to GND using BLACK; record initials ____.

Unpowered test

Instruction / kind / points / expected	Evidence
Meter divider and adapter · resistance · VIN–UV_SENSE–GND, VIN–OV_SENSE–GND, U3 pins · 150 kΩ/10 kΩ and 255 kΩ/10 kΩ networks; pin map exact	
Continuity-map adapter header · continuity · adapter pins 1–6 to header · one-to-one recorded mapping before installation	

Powered test

Instruction / input / output / evidence / limits	Measured
Sweep protected VIN · VIN · TREAD_OK · record low/high transitions · low below UV, high at 8.00 V, low above OV; UV rising 6.25–6.55 V; OV falling 10.30–10.90 V	

STOP gate: TREAD_OK three-state behavior or thresholds fail; never exceed protected VIN about 11.1 V; stop if TVS draws appreciable current or warms.

PASS gate: Both sweep directions and GPIO6 isolated observation match limits. **PASS — continue** to Cluster 5 only.

Cluster 5 — AHC08 permission logic

Dependencies: Cluster 3, Cluster 4
Inputs: LOGIC_3V3, TREAD_OK, RELAY_CMD, TX_ENABLE, GND

Outputs: RELAY_GATE, TX_GATE
Source state: USB logic and bench VIN; remove the corresponding DevKit command jumper before manual injection

Source state record: USB logic and bench VIN; remove the corresponding DevKit command jumper before manual injection

Parts

U4 SN74AHC08N PDIP-14 · 296-4524-5-ND · notch verified, across center channel. **C5** 100 nF · BC1084CT-ND · non-polar. **R1/R2** 10 kΩ command pull-downs · NOT ORDERED.

Operator part record: U4/C5/R1/R2 · SN74AHC08N; capacitor; pull-downs · PDIP-14; 100 nF; two 10 kΩ · 296-4524-5-ND; BC1084CT-ND; NOT ORDERED · notch verified; capacitor non-polar

Operator wiring record: See numbered point-to-point steps below.

Build — exact pin/net/color wiring

Step	Ref	Color	From	To	Part	Note
1	58	BLUE	LOGIC_3V3	AHC08 pin 14	Wire	VCC Connect U4 pin 14 to LOGIC_3V3 using BLUE; record initials ____.
2	59	BLACK	AHC08 pin 7	GND	Wire	Ground Connect U4 pin 7 to GND using BLACK; record initials ____.
3	60	YELLOW	DevKit GPIO21	AHC08 pin 1	Wire	RELAY_CMD Connect GPIO21 jumper post endpoint to U4 pin 1 / RELAY_CMD using YELLOW; record initials ____.
4	63	YELLOW	TREAD_OK	AHC08 pin 2	Wire	Relay permission Connect U4 pin 2 to TREAD_OK using YELLOW; record initials ____.
5	64	YELLOW	AHC08 pin 3	RELAY_GATE	Wire	Relay gate output Connect U4 pin 3 to RELAY_GATE using YELLOW; record initials ____.
6	65	YELLOW	DevKit GPIO15	AHC08 pin 4	Wire	TX_ENABLE Connect GPIO15 jumper post endpoint to U4 pin 4 / TX_ENABLE using YELLOW; record initials ____.
7	68	YELLOW	TREAD_OK	AHC08 pin 5	Wire	TX permission Connect U4 pin 5 to TREAD_OK using YELLOW; record initials ____.
8	69	YELLOW	AHC08 pin 6	TX_GATE	Wire	TX gate output Connect U4 pin 6 to TX_GATE using YELLOW; record initials ____.
9	70a	BLACK	AHC08 pin 9	GND	Wire	Tie unused inputs low legacy Ref 70; 4 wires Connect U4 9 to GND using BLACK; record initials ____.
10	70b	BLACK	AHC08 pin 10	GND	Wire	Tie unused inputs low legacy Ref 70; 4 wires Connect U4 10 to GND using BLACK; record initials ____.
11	70c	BLACK	AHC08 pin 12	GND	Wire	Tie unused inputs low legacy Ref 70; 4 wires Connect U4 12 to GND using BLACK; record initials ____.
12	70d	BLACK	AHC08 pin 13	GND	Wire	Tie unused inputs low legacy Ref 70; 4 wires Connect U4 13 to GND using BLACK; record initials ____.
13	71a	NO WIRE	AHC08 pin 8	NO CONNECTION	NO WIRE	Unused outputs legacy Ref 71; 2 intentional opens Leave U4 pin 8 at NC using NO WIRE; record initials ____.
14	71b	NO WIRE	AHC08 pin 11	NO CONNECTION	NO WIRE	Unused outputs legacy Ref 71; 2 intentional opens Leave U4 pin 11 at NC using NO WIRE; record initials ____.
15	61	YELLOW	DevKit GPIO21	R_RELAY_CMD_PD 10 kΩ, end 1	Wire	Command pull-down DevKit GPIO21 ↔ removable GPIO21 jumper ↔ RELAY_CMD / AHC08 pin 1. Connect GPIO21 jumper post endpoint to U4 pin 1 / RELAY_CMD using YELLOW; record initials ____. Connect R_RELAY_CMD_PD 1 to RELAY_CMD using YELLOW; record initials ____.
16	62	BLACK	R_RELAY_CMD_PD end 2	GND	Wire	Pull-down return Connect R_RELAY_CMD_PD 2 to GND using BLACK; record initials ____.
17	66	YELLOW	DevKit GPIO15	R_TX_ENABLE_PD 10 kΩ, end 1	Wire	Enable pull-down DevKit GPIO15 ↔ removable GPIO15 jumper ↔ TX_ENABLE / AHC08 pin 4. Connect GPIO15 jumper post endpoint to U4 pin 4 / TX_ENABLE using YELLOW; record initials ____. Connect R_TX_ENABLE_PD 1 to TX_ENABLE using YELLOW; record initials ____.
18	67	BLACK	R_TX_ENABLE_PD end 2	GND	Wire	Pull-down return Connect R_TX_ENABLE_PD 2 to GND using BLACK; record initials ____.
19	72	BLUE	AHC08 pin 14	C_AHC08 100 nF, end 1	Wire	Place close to IC Connect U4 pin 14 to LOGIC_3V3 using BLUE; record initials ____. Connect C_AHC08 1 to LOGIC_3V3 using BLUE; record initials ____.
20	73	BLACK	C_AHC08 end 2	AHC08 pin 7	Wire	Decoupling return Connect U4 pin 7 to GND using BLACK; record initials ____. Connect C_AHC08 2 to GND using BLACK; record initials ____.

Unpowered test

Instruction / kind / points / expected	Evidence
Check DIP orientation and control mapping · mapping · U4 pins 1–14 and named nets · exact map, unused inputs grounded, outputs not shorted	

Powered test

Instruction / input / output / evidence / limits	Measured
Exercise both AND gates · RELAY_CMD/TX_ENABLE plus TREAD_OK · RELAY_GATE/TX_GATE · record 00,10,01,11 truth tables · output high only for command=1 and TREAD_OK=1	

Remove GPIO15 jumper before manual injection of TX_ENABLE. Remove GPIO21 jumper before manual injection of RELAY_CMD. Never drive against a connected DevKit GPIO.

STOP gate: Any gate rises without both inputs, a command GPIO remains connected during manual drive, or supply/reset behavior is abnormal.

PASS gate: Both four-state truth tables pass and temporary drives are removed. **PASS — continue** to Cluster 6 only.

Operator: _____ Date: _____ Signed PASS: _____
Cluster 5 · AHC08 permissions

Cluster 6 — TPS709 and BC337 driver

Dependencies: Cluster 1, Cluster 5
Inputs: VIN, RELAY_GATE, GND

Outputs: +5V_RLY, RELAY_COIL-
Source state: Valid VIN/TREAD_OK; COIL POWER remains removed for the unloaded test

Parts

U5 TPS70950DBVR · 296-35484-1-ND; **adapter** LCQT-SOT23-6 · A879AR-ND; **header** cut from PRPC040SAAN-RC · S1011EC-40-ND; **Q1** BC337-40 · 4878-BC337-40CT-ND; **RBASE** 560 Ω · MFR-25FBF52-560R-ND; 10 kΩ · NOT ORDERED; 1 μF · 445-173583-1-ND; 4.7 μF · 732-8660-1-ND; **JP2** COIL POWER · NOT ORDERED.

Operator part record: U5/Q1/RBASE/caps/JP2 · TPS70950DBVR; BC337-40; 560 Ω/10 kΩ; 1 μF/4.7 μF; COIL POWER · 5 V regulator and unloaded driver · 296-35484-1-ND; A879AR-ND; 4878-BC337-40CT-ND; MFR-25FBF52-560R-ND; 445-173583-1-ND; 732-8660-1-ND; NOT ORDERED · adapter and Q1 leads mapped; electrolytic negative to GND; link removed

Operator wiring record: See numbered point-to-point steps below.

Operator header record: HDR5 · PRPC040SAAN-RC header · cut header · S1011EC-40-ND · map before installation

Build — exact pin/net/color wiring

Step	Ref	Color	From	To	Part	Note
1	74	ORANGE	VIN	TPS709 pin 1 IN	Wire	Regulator input Connect U5 pin 1 to VIN using ORANGE; record initials ____.
2	75	BLACK	TPS709 pin 2	GND	Wire	Ground Connect U5 pin 2 to GND using BLACK; record initials ____.
3	76	YELLOW	RELAY_GATE	TPS709 pin 3 EN	Wire	Hardware-gated enable Connect U5 pin 3 to RELAY_GATE using YELLOW; record initials ____.
4	78	VIOLET	TPS709 pin 5 OUT	+5V_RLY	Wire	Relay supply Connect U5 pin 5 to +5V_RLY using VIOLET; record initials ____.
5	77	NO WIRE	TPS709 pin 4	NO CONNECTION	—	NC pin Leave U5 pin 4 at NC using NO WIRE; record initials ____.
6	79	ORANGE	VIN	C_TPS709_IN 1 μF, end 1	Wire	Input bypass Connect C_TPS709_IN 1 to VIN using ORANGE; record initials ____.
7	80	BLACK	C_TPS709_IN end 2	GND	Wire	Bypass return Connect C_TPS709_IN 2 to GND using BLACK; record initials ____.
8	81	VIOLET	+5V_RLY	C_TPS709_OUT 4.7 μF positive	Wire	Output capacitor Connect C_TPS709_OUT positive to +5V_RLY using VIOLET; record initials ____.
9	82	BLACK	C_TPS709_OUT negative	GND	Wire	Output return Connect C_TPS709_OUT negative to GND using BLACK; record initials ____.
10	83	YELLOW	RELAY_GATE	R_BASE 560 Ω, end 1	Wire	Base resistor Connect R_BASE 1 to RELAY_GATE using YELLOW; record initials ____.
11	84	YELLOW	R_BASE end 2	BC337 base	Wire	Verify physical lead Connect R_BASE 2 to Q_BASE using YELLOW; record initials ____. Connect Q1 base to Q_BASE using YELLOW; record initials ____.
12	85	YELLOW	BC337 base	R_BASE_PD 10 kΩ, end 1	Wire	Base pull-down Connect Q1 base to Q_BASE using YELLOW; record initials ____. Connect R_BASE_PD 1 to Q_BASE using YELLOW; record initials ____.
13	86	BLACK	R_BASE_PD end 2	GND	Wire	Pull-down return Connect R_BASE_PD 2 to GND using BLACK; record initials ____.
14	87	BLACK	BC337 emitter	GND	Wire	Verify physical lead Connect Q1 emitter to GND using BLACK; record initials ____.
15	NEW C6-16	VIOLET	Q1 collector	RELAY_COIL-	Wire	Connect Q1 collector to RELAY_COIL- using VIOLET; record initials ____.
16	NEW C6-17	VIOLET	JP2 supply side	+5V_RLY	Wire	Connect JP2 supply side to +5V_RLY using VIOLET; record initials ____.
17	NEW C6-18	VIOLET	JP2 coil side	RELAY_COIL+	Wire	Connect JP2 coil side to RELAY_COIL+ using VIOLET; record initials ____.

Unpowered test

Instruction / kind / points / expected	Evidence
Verify adapter, transistor leads, and isolation · polarity · U5 pins, Q1 B/C/E, output capacitor, JP2 · exact map; COIL POWER open	

Powered test

Instruction / input / output / evidence / limits	Measured
Test gate unloaded · RELAY_GATE · +5V_RLY/Q1 base · record on/off · TPS709 enabled 4.75–5.25 V; TPS709 disabled below 0.25 V	

STOP gate: COIL POWER is installed, enabled or disabled limits fail, output does not discharge below 0.25 V, or Q1/U5 heats.

PASS gate: Unloaded regulator and base drive pass with coil isolated. **PASS — continue** to Cluster 7 only.

Operator: _____

Date: _____

Signed PASS: _____

Cluster 7 — Relay coil, local contacts, and feedback

Dependencies: Cluster 6, Cluster 3
Inputs: +5V_RLY, RELAY_COIL–, LOGIC_3V3, GND

Outputs: LOCAL_NC, LOCAL_NO, FB_NC, FB_NO
Source state: Local-only bench exercise; treadmill/RJ45 cables absent; bounded relay exerciser verified; loaded relay limit no more than 500 mA

Parts

K1 G5V-2 DC5 DPDT · 39-G5V-2DC5-ND · bottom-view drawing, meter-map physical pins. **D3** P6KE6.8CA bidirectional · 18-P6KE6.8CACT-ND. **JP2** COIL POWER · NOT ORDERED. **RFB** two 10 k Ω · NOT ORDERED.

Operator part record: K1/D3/JP2/RFB · G5V-2 DC5; P6KE6.8CA; COIL POWER; pull-ups · DPDT 5 V; bidirectional TVS; removable; two 10 k Ω · 39-G5V-2DC5-ND; 18-P6KE6.8CACT-ND; NOT ORDERED · bottom-view drawing; meter-map physical pins; TVS bidirectional

Operator wiring record: See numbered point-to-point steps below.

Build — exact pin/net/color wiring

Step	Ref	Color	From	To	Part	Note
1	89	VIOLET	+5V_RLY	K1 pin 1	COIL POWER jumper	Leave removed initially Connect K1 pin 1 to RELAY_COIL+ using VIOLET; record initials ____.
2	88	VIOLET	BC337 collector	K1 pin 16	Wire	RELAY_COIL– Connect K1 pin 16 to RELAY_COIL– using VIOLET; record initials ____.
3	90	VIOLET	K1 pin 1	P6KE6.8CA lead 1	Wire	Bidirectional coil TVS Connect K1 pin 1 to RELAY_COIL+ using VIOLET; record initials _____. Connect D3 lead 1 to RELAY_COIL+ using VIOLET; record initials ____.
4	91	VIOLET	P6KE6.8CA lead 2	K1 pin 16	Wire	Bidirectional coil TVS Connect K1 pin 16 to RELAY_COIL– using VIOLET; record initials _____. Connect D3 lead 2 to RELAY_COIL– using VIOLET; record initials ____.
5	NEW C7-5	WHITE	K1 pin 4	LOCAL_NC	Wire	Connect K1 pin 4 to LOCAL_NC using WHITE; record initials ____.
6	NEW C7-6	WHITE	K1 pin 6	LOCAL_TX_SELECTED	Wire	Connect K1 pin 6 to LOCAL_TX_SELECTED using WHITE; record initials ____.
7	NEW C7-7	WHITE	K1 pin 8	LOCAL_NO	Wire	Connect K1 pin 8 to LOCAL_NO using WHITE; record initials ____.
8	105	BLACK	K1 pin 11	GND	Wire	Feedback COM Connect K1 pin 11 to GND using BLACK; record initials ____.
9	106	GREEN	K1 pin 13	DevKit GPIO4	Wire	NC feedback Connect K1 pin 13 to K1_NC_FB using GREEN; record initials _____. Connect DEVKIT GPIO4 to K1_NC_FB using GREEN; record initials ____.
10	109	GREEN	K1 pin 9	DevKit GPIO5	Wire	NO feedback Connect K1 pin 9 to K1_NO_FB using GREEN; record initials _____. Connect DEVKIT GPIO5 to K1_NO_FB using GREEN; record initials ____.
11	107	BLUE	LOGIC_3V3	R_FB_NC 10 k Ω , end 1	Wire	NC pull-up Connect R_FB_NC 1 to LOGIC_3V3 using BLUE; record initials ____.
12	108	GREEN	R_FB_NC end 2	DevKit GPIO4	Wire	NC pull-up node Connect R_FB_NC 2 to K1_NC_FB using GREEN; record initials _____. Connect DEVKIT GPIO4 to K1_NC_FB using GREEN; record initials ____.
13	110	BLUE	LOGIC_3V3	R_FB_NO 10 k Ω , end 1	Wire	NO pull-up Connect R_FB_NO 1 to LOGIC_3V3 using BLUE; record initials ____.
14	111	GREEN	R_FB_NO end 2	DevKit GPIO5	Wire	NO pull-up node Connect R_FB_NO 2 to K1_NO_FB using GREEN; record initials _____. Connect DEVKIT GPIO5 to K1_NO_FB using GREEN; record initials ____.

Unpowered test

Instruction / kind / points / expected	Evidence
Map relay before attaching local fixture · continuity · pins 1–16, 4–6, 8–6, 13–11, 9–11 · coil about 50 Ω ; NC pairs closed; NO pairs open	

Powered test

Instruction / input / output / evidence / limits	Measured
Exercise bounded relay and fail sources · +5V_RLY/RELAY_COIL– · local contacts/feedback · record current, voltage, VCE, timing, temperature · coil 90–110 mA, voltage at least 4.50 V, BC337 VCE no more than 0.30 V; (1,0) energized, (0,1) bypass; 00 or 11 fault	

Bound operator part record: K1/D3/JP2/RFB · G5V-2 DC5; P6KE6.8CA; COIL POWER; pull-ups · DPDT 5 V; bidirectional TVS; removable; two 10 k Ω · 39-G5V-2DC5-ND; 18-P6KE6.8CACT-ND; NOT ORDERED · bottom-view drawing; meter-map physical pins; TVS bidirectional

Relay exerciser identity: _____ **Firmware identity:** _____

From energized state, removal of USB logic power must release to NC bypass; removal of GPIO21 command must release to NC bypass; removal of TREAD_OK must release to NC bypass; removal of VIN must release to NC bypass. Each NC bypass return must occur in no more than 100 ms. Then hold five minutes: TPS709 and BC337 at or below 45°C and no more than 10°C over ambient.

Dedicated fail-release and thermal evidence

Evidence field	Measured	Limit
USB logic release (ms)		NC bypass $\leq$ 100 ms
GPIO21 release (ms)		NC bypass $\leq$ 100 ms
TREAD_OK release (ms)		NC bypass $\leq$ 100 ms
VIN release (ms)		NC bypass $\leq$ 100 ms
Ambient temperature (°C)		Record before hold
TPS709 temperature (°C)		$\leq$ 45°C and $\leq$ 10°C over ambient
BC337 temperature (°C)		$\leq$ 45°C and $\leq$ 10°C over ambient

STOP gate: Any electrical, feedback, release, reset, or five-minute thermal gate fails. Cluster 7 is local-only; it makes no end-to-end RJ45 claim.

PASS gate: Local contacts, feedback, current, voltage, release, and thermal evidence all pass. **PASS — continue** to Cluster 8 only.

Operator:

Date:

Signed PASS:

Cluster 8 — AHC126 and UART taps

Dependencies: Cluster 3, Cluster 5, Cluster 7
Inputs: LOGIC_3V3, TX_GATE, GPIO17_TX, LOCAL_NC, LOCAL_NO, GND

Outputs: TX_DRV, UART_BUFFER_HIGH_Z, UART_BUFFER_FOLLOWS, GPIO18_RX_LOCAL, GPIO16_RX_LOCAL, LOCAL_TX_SELECTED
Source state: Local future-interface fixture only; treadmill and both RJ45 breakouts absent; bounded UART/relay exerciser identity recorded

Parts

U6 SN74AHC126N PDIP-14 · 296-4535-5-ND · notch verified. **C6** 100 nF · BC1084CT-ND. **R3** 100 Ω and **R4/R5** 10 k Ω · NOT ORDERED.

Operator part record: U6/C6/R3/R4/R5 · SN74AHC126N; capacitor; series/tap resistors · PDIP-14; 100 nF; 100 Ω ; two 10 k Ω · 296-4535-5-ND; BC1084CT-ND; NOT ORDERED · notch verified; capacitor non-polar

Operator wiring record: See numbered point-to-point steps below.

Build — exact pin/net/color wiring

Step	Ref	Color	From	To	Part	Note
1	92	BLUE	LOGIC_3V3	AHC126 pin 14	Wire	VCC Connect U6 pin 14 to LOGIC_3V3 using BLUE; record initials ____.
2	93	BLACK	AHC126 pin 7	GND	Wire	Ground Connect U6 pin 7 to GND using BLACK; record initials ____.
3	94	YELLOW	TX_GATE	AHC126 pin 1	Wire	Gate 1 OE Connect U6 pin 1 to TX_GATE using YELLOW; record initials ____.
4	95	YELLOW	DevKit GPIO17	AHC126 pin 2	Wire	Gate 1 input Connect U6 pin 2 to GPIO17_TX using YELLOW; record initials ____.
5	96	YELLOW	AHC126 pin 3	R_TX 100 Ω , end 1	Wire	Gate 1 output Connect U6 pin 3 to TX_BUF using YELLOW; record initials _____. Connect R_TX 1 to TX_BUF using YELLOW; record initials ____.
6	NEW C8-7	YELLOW	R_TX 2	TX_DRV	Wire	Connect R_TX 2 to TX_DRV using YELLOW; record initials ____.
7	97a	BLACK	AHC126 pin 4	GND	Wire	Unused enables/inputs low legacy Ref 97; 6 wires Connect U6 4 to GND using BLACK; record initials ____.
8	97b	BLACK	AHC126 pin 5	GND	Wire	Unused enables/inputs low legacy Ref 97; 6 wires Connect U6 5 to GND using BLACK; record initials ____.
9	97c	BLACK	AHC126 pin 9	GND	Wire	Unused enables/inputs low legacy Ref 97; 6 wires Connect U6 9 to GND using BLACK; record initials ____.
10	97d	BLACK	AHC126 pin 10	GND	Wire	Unused enables/inputs low legacy Ref 97; 6 wires Connect U6 10 to GND using BLACK; record initials ____.
11	97e	BLACK	AHC126 pin 12	GND	Wire	Unused enables/inputs low legacy Ref 97; 6 wires Connect U6 12 to GND using BLACK; record initials ____.
12	97f	BLACK	AHC126 pin 13	GND	Wire	Unused enables/inputs low legacy Ref 97; 6 wires Connect U6 13 to GND using BLACK; record initials ____.
13	98a	NO WIRE	AHC126 pin 6	NO CONNECTION	NO WIRE	Unused outputs legacy Ref 98; 3 intentional opens Leave U6 pin 6 at NC using NO WIRE; record initials ____.
14	98b	NO WIRE	AHC126 pin 8	NO CONNECTION	NO WIRE	Unused outputs legacy Ref 98; 3 intentional opens Leave U6 pin 8 at NC using NO WIRE; record initials ____.
15	98c	NO WIRE	AHC126 pin 11	NO CONNECTION	NO WIRE	Unused outputs legacy Ref 98; 3 intentional opens Leave U6 pin 11 at NC using NO WIRE; record initials ____.
16	99	BLUE	AHC126 pin 14	C_AHC126 100 nF, end 1	Wire	Place close to IC Connect U6 pin 14 to LOGIC_3V3 using BLUE; record initials _____. Connect C_AHC126 1 to LOGIC_3V3 using BLUE; record initials ____.
17	100	BLACK	C_AHC126 end 2	AHC126 pin 7	Wire	Decoupling return Connect U6 pin 7 to GND using BLACK; record initials _____. Connect C_AHC126 2 to GND using BLACK; record initials ____.
18	NEW C8-19	WHITE	R_CONS_RX 1	LOCAL_CONSOLE_6_STUB	Wire	Connect R_CONS_RX 1 to LOCAL_CONSOLE_6_STUB using WHITE; record initials ____.
19	NEW C8-20	GREEN	R_CONS_RX 2	GPIO18_RX_LOCAL	Wire	Connect R_CONS_RX 2 to GPIO18_RX_LOCAL using GREEN; record initials ____.
20	NEW C8-21	GREEN	DEVKIT GPIO18	GPIO18_RX_LOCAL	Wire	Connect DEVKIT GPIO18 to GPIO18_RX_LOCAL using GREEN; record initials ____.
21	NEW C8-22	WHITE	R_PIN3_RX 1	LOCAL_PIN3_STUB	Wire	Connect R_PIN3_RX 1 to LOCAL_PIN3_STUB using WHITE; record initials ____.
22	NEW C8-23	GREEN	R_PIN3_RX 2	GPIO16_RX_LOCAL	Wire	Connect R_PIN3_RX 2 to GPIO16_RX_LOCAL using GREEN; record initials ____.
23	NEW C8-24	GREEN	DEVKIT GPIO16	GPIO16_RX_LOCAL	Wire	Connect DEVKIT GPIO16 to GPIO16_RX_LOCAL using GREEN; record initials ____.
24	NEW C8-25	WHITE	K1 pin 4 / LOCAL_NC	LOCAL_CONSOLE_6_STUB	Wire	Connect K1 pin 4 / LOCAL_NC to LOCAL_CONSOLE_6_STUB using WHITE; record initials ____.
25	NEW C8-26	YELLOW	K1 pin 8 / LOCAL_NO	TX_DRV	Wire	Connect K1 pin 8 / LOCAL_NO to TX_DRV using YELLOW; record initials ____.
26	NEW C8-27	YELLOW	K1 pin 6	LOCAL_TX_SELECTED	Wire	Connect K1 pin 6 to LOCAL_TX_SELECTED using YELLOW; record initials ____.

Unpowered test

Instruction / kind / points / expected	Evidence
Check local buffer map · continuity · U6 pins, 100 Ω resistor, local TX_DRV test point · exact map; disabled output isolated; no connector-side wiring	
Verify GPIO18 local receive tap · resistance · LOCAL_CONSOLE_6_STUB → GPIO18_RX_LOCAL · 10 kΩ nominal (9–11 kΩ); isolated from LOGIC_3V3, GND, and TX_DRV	
Verify GPIO16 local receive tap · resistance · LOCAL_PIN3_STUB → GPIO16_RX_LOCAL · 10 kΩ nominal (9–11 kΩ); isolated from LOGIC_3V3, GND, and TX_DRV	

Powered test

Instruction / input / output / evidence / limits	Measured
Exercise local buffer enable · TX_GATE/GPIO17_TX/local fixture · TX_DRV/UART_BUFFER_HIGH_Z/UART_BUFFER_FOLLOWS · capture logic traces · pin 3 high impedance disabled and follows pin 2 enabled	
Stimulate GPIO18 local receive tap · LOCAL_CONSOLE_6_STUB low/high from bounded UART exerciser · GPIO18_RX_LOCAL · record low/high voltage and observed bits · low ≤0.40 V; high ≥2.40 V; bits follow	
Stimulate GPIO16 local receive tap · LOCAL_PIN3_STUB low/high from bounded UART exerciser · GPIO16_RX_LOCAL · record low/high voltage and observed bits · low ≤0.40 V; high ≥2.40 V; bits follow	
Exercise bounded local relay transfer · TX_DRV/K1 LOCAL_NC/LOCAL_NO with bounded relay exerciser · LOCAL_TX_SELECTED · record NC/NO selected resistance and TX waveform · selected path below 2 Ω; unselected path open; waveform follows TX_DRV	

Relay exerciser identity: _____ Firmware identity: _____

Dedicated local future-interface evidence

Evidence field	Measured	Limit/state
GPIO18_RX_LOCAL low (V)		≤0.40 V
GPIO18_RX_LOCAL high (V)		≥2.40 V
GPIO16_RX_LOCAL low (V)		≤0.40 V
GPIO16_RX_LOCAL high (V)		≥2.40 V
LOCAL_TX_SELECTED NC (Ω)		<2 Ω selected
LOCAL_TX_SELECTED NO (Ω)		<2 Ω selected

STOP gate: TX_DRV drives while disabled, enabled waveform does not follow, either receive tap fails, local relay selection fails, any connector-side wire is installed, or exerciser identity is absent.

PASS gate: Local high impedance, following, receive taps, and relay-selected TX pass; connector-side checks remain deferred. **PASS — continue** to Cluster 9 only.

Operator: _____ Date: _____ Signed PASS: _____
Cluster 8 · UART isolation and observation

Cluster 9 — Indicators and VBUS sensing

Dependencies: Cluster 3
Inputs: LOGIC_3V3, GPIO38, USB_5V, GND

Outputs: POWER_LED, STATUS_LED, VBUS_PRESENT_N
Source state: USB-only test then standalone-only test; never join VBUS to a local rail

Parts

Q2 2N7000 · 4878-2N7000CT-ND; **D4** green LTL-4233 · 160-1130-ND; **D5** red 151051RS11000 · 732-5016-ND; 330 Ω , 1 k Ω , and 10 k Ω resistors · NOT ORDERED; 100 nF · BC1084CT-ND.

Operator part record: Q2/D4/D5/resistors/C7 · 2N7000; green/red LEDs; current/sense resistors; capacitor · MOSFET; 330 Ω /1 k Ω /10 k Ω ; 100 nF · 4878-2N7000CT-ND; 160-1130-ND; 732-5016-ND; BC1084CT-ND; NOT ORDERED · Q2 G/D/S mapped; LED long lead anode; flat side cathode

Operator wiring record: See numbered point-to-point steps below.

Operator bypass wiring record: C7 · leads 1/2 · LOGIC_3V3/GND · BLUE/BLACK

Build — exact pin/net/color wiring

Step	Ref	Color	From	To	Part	Note
1	112	VIOLET	DevKit 5V/VBUS	2N7000 gate	Wire	Verify physical lead Connect DEVKIT 5V/VBUS to VBUS_SENSE using VIOLET; record initials _____. Connect Q2 gate to VBUS_SENSE using VIOLET; record initials _____. .
2	113	VIOLET	DevKit 5V/VBUS	R_VBUS_DISCHARGE 10 k Ω , end 1	Wire	Gate discharge Connect DEVKIT 5V/VBUS to VBUS_SENSE using VIOLET; record initials _____. Connect R_VBUS_DISCHARGE 1 to VBUS_SENSE using VIOLET; record initials _____. .
3	114	BLACK	R_VBUS_DISCHARGE end 2	GND	Wire	Discharge return Connect R_VBUS_DISCHARGE 2 to GND using BLACK; record initials _____. .
4	115	BLACK	2N7000 source	GND	Wire	Verify physical lead Connect Q2 source to GND using BLACK; record initials _____. .
5	116	GREEN	2N7000 drain	DevKit GPIO7	Wire	VBUS_PRESENT_N Connect Q2 drain to VBUS_PRESENT_N using GREEN; record initials _____. Connect DEVKIT GPIO7 to VBUS_PRESENT_N using GREEN; record initials _____. .
6	117	BLUE	LOGIC_3V3	R_VBUS_PULLUP 10 k Ω , end 1	Wire	Drain pull-up Connect R_VBUS_PULLUP 1 to LOGIC_3V3 using BLUE; record initials _____. .
7	118	GREEN	R_VBUS_PULLUP end 2	DevKit GPIO7	Wire	Pull-up node Connect R_VBUS_PULLUP 2 to VBUS_PRESENT_N using GREEN; record initials _____. Connect DEVKIT GPIO7 to VBUS_PRESENT_N using GREEN; record initials _____. .
8	119	YELLOW	DevKit GPIO38	R_STATUS 330 Ω , end 1	Wire	Status LED Connect DEVKIT GPIO38 to STATUS_LED using YELLOW; record initials _____. Connect R_STATUS 1 to STATUS_LED using YELLOW; record initials _____. .
9	120	YELLOW	R_STATUS end 2	Green LED anode	Wire	Long LED lead Connect R_STATUS 2 to LED_GREEN_A using YELLOW; record initials _____. Connect LED_GREEN anode to LED_GREEN_A using YELLOW; record initials _____. .
10	121	BLACK	Green LED cathode	GND	Wire	Short/flat-side lead Connect LED_GREEN cathode to GND using BLACK; record initials _____. .
11	122	BLUE	LOGIC_3V3	R_POWER_LED 1 k Ω , end 1	Wire	Power LED Connect R_POWER_LED 1 to LOGIC_3V3 using BLUE; record initials _____. .
12	123	YELLOW	R_POWER_LED end 2	Red LED anode	Wire	Long LED lead Connect R_POWER_LED 2 to LED_RED_A using BLUE; record initials _____. Connect LED_RED anode to LED_RED_A using BLUE; record initials _____. .
13	124	BLACK	Red LED cathode	GND	Wire	Short/flat-side lead Connect LED_RED cathode to GND using BLACK; record initials _____. .
14	125	BLUE	LOGIC_3V3	C_LOGIC 100 nF, end 1	Wire	Logic rail bypass Connect C_LOGIC 1 to LOGIC_3V3 using BLUE; record initials _____. .
15	126	BLACK	C_LOGIC end 2	GND	Wire	Bypass return C7 lead 1 to LOGIC_3V3 and C7 lead 2 to GND. Connect C_LOGIC 2 to GND using BLACK; record initials _____. .

Unpowered test

Instruction / kind / points / expected	Evidence
Verify transistor leads and LED polarity · polarity · Q2 G/D/S, both LED anodes/cathodes · exact map; VBUS isolated from all local positive rails	

Powered test

Instruction / input / output / evidence / limits	Measured
Toggle source and status · USB_5V/LOGIC_3V3/GPIO38 · VBUS_PRESENT_N/LEDs · record truth/voltage · VBUS sense active-low; USB present low, USB absent high; red power and green status indications correct	

Bound operator part record: Q2/D4/D5/resistors/C7 · 2N7000; green/red LEDs; current/sense resistors; capacitor · MOSFET; 330 Ω /1 k Ω /10 k Ω ; 100 nF · 4878-2N7000CT-ND; 160-1130-ND; 732-5016-ND; BC1084CT-ND; NOT ORDERED · Q2 G/D/S mapped; LED long lead anode; flat side cathode

VBUS sense is active-low and isolated: no connection may ever join VBUS to a local rail.

STOP gate: VBUS joins a local rail, active-low truth is wrong, LED polarity/current limiting is wrong, or sources overlap.

PASS gate: Both LEDs and isolated active-low VBUS sense pass under exclusive sources. **PASS — continue** to Cluster 10 only.

Operator:

Date:

Signed PASS:

Cluster 10 — Whole-device standalone bench test

Dependencies: Cluster 1, Cluster 2, Cluster 3, Cluster 4, Cluster 5, Cluster 6, Cluster 7, Cluster 8, Cluster 9
Inputs: +8V_RAW, GND, observer firmware

Outputs: standalone boot, Wi-Fi observation, event log, UART idle evidence, verified observer state
Source state: USB physically absent; STANDALONE POWER installed; observer verified; relay off; TX disabled

Outputs record: standalone boot · Wi-Fi observation · event log · UART idle evidence

Parts

ASSEMBLY completed Clusters 1–9 · NOT ORDERED · inspect all removable links and polarities. **Firmware artifact** observer image and manifest · NOT ORDERED · exact immutable identity required.

Operator part record: ASSEMBLY/FW · completed Clusters 1–9; observer image and manifest · whole device; immutable diagnostic artifact · NOT ORDERED · inspect links/polarities; observer manifest proves relay and TX disabled

Operator wiring record: USB · plug · DISCONNECTED · NO WIRE; STANDALONE POWER · between · TSR_3V3 and LOGIC_3V3 · BLUE; bench · positive lead · +8V_RAW · RED; bench · return lead · GND · BLACK

Build — exact pin/net/color wiring

Step	Ref	Color	From	To	Part	Note
1	NEW C10-1	NO WIRE	USB plug	DISCONNECTED	NO WIRE	Connect USB plug to DISCONNECTED using NO WIRE.
2	NEW C10-2	BLUE	STANDALONE POWER between	TSR_3V3 and LOGIC_3V3	Jumper	Install STANDALONE POWER between TSR_3V3 and LOGIC_3V3 using BLUE.
3	NEW C10-3	RED	bench positive lead	+8V_RAW	Bench lead	Connect the bench positive lead to +8V_RAW using RED.
4	NEW C10-4	BLACK	bench return lead	GND	Bench lead	Connect the bench return lead to GND using BLACK.

Unpowered test

Instruction / kind / points / expected	Evidence
Final standalone source audit · resistance · +8V_RAW/VIN/TSR_3V3/LOGIC_3V3/+5V_RLY to GND · no shorts; USB absent; source handoff exact	

Powered test

Instruction / input / output / evidence / limits	Measured
Boot observer from raw input · +8V_RAW · standalone boot/Wi-Fi/event log/UART idle · record rails, reset log, GPIO16/GPIO18 idle · VIN 7.20–7.90 V; TSR_3V3 and LOGIC_3V3 3.20–3.40 V; relay off, feedback (0,1), TX disabled	

Firmware identity: _____ **Observer identity:** _____

Observer manifest: _____ (path or SHA proving relay and TX outputs disabled)

Dedicated UART idle evidence

Evidence field	Measured	State
GPIO16 UART idle (V)		Before UART configuration
GPIO18 UART idle (V)		Before UART configuration

STOP gate: Identity/manifest missing, rail out of range, reset, Wi-Fi/event-log failure, UART idle missing, relay enables, TX enables, or feedback differs from (0,1).

PASS gate: Identified observer boots standalone and all passive observations pass. **PASS — continue** to Cluster 11 only.

Operator: _____ Date: _____ Signed PASS: _____

Cluster 11 — RJ45 pass-through and treadmill bypass

Cluster 11 is the end-to-end CONSOLE.6 ↔ MOTOR.6 NC bypass proof.

Dependencies: Cluster 7, Cluster 8, Cluster 10
Inputs: CONSOLE.1–8, MOTOR.1–8, verified observer state

Outputs: eight mapped conductors, end-to-end CONSOLE.6 ↔ MOTOR.6 NC bypass, bypass-only treadmill evidence
Source state: USB absent; standalone installed; observer verified; relay off; TX disabled; source → state → harness → direct-path sequence

Parts

J1/J2 CONSOLE/MOTOR RJ45 breakouts · NOT ORDERED · meter-map terminal numbers. **Harness** reviewed dual-pin fused-DMM current harness · NOT ORDERED. **K1** verified Cluster 7 relay contacts · 39-G5V-2DC5-ND.

Operator part record: J1/J2/HARNESS/K1 · CONSOLE/MOTOR RJ45 breakouts; dual-pin fused-DMM harness; verified relay · two 8P8C breakouts; reviewed current harness; G5V-2 · NOT ORDERED; 39-G5V-2DC5-ND · meter-map connector terminals; verify fused jack/polarity; relay bottom-view map proven
Operator wiring record: J1/J2 · terminals · unwired · NO WIRE; J1 · pin 1 · GND · BLACK; J2 · pin 1 · GND · BLACK; J1 · pin 7 · GND · BLACK; J2 · pin 7 · GND · BLACK; J1 · pin 2 · +8V_RAW · RED; J2 · pin 2 · +8V_RAW · RED; J1 · pin 8 · +8V_RAW · RED; J2 · pin 8 · +8V_RAW · RED; CONSOLE.3 · endpoint · MOTOR.3 · WHITE; CONSOLE.4 · endpoint · MOTOR.4 · WHITE; CONSOLE.5 · endpoint · MOTOR.5 · WHITE; CONSOLE.6 · endpoint · K1 pin 4 · YELLOW; MOTOR.6 · endpoint · K1 pin 6 · YELLOW; K1 · pin 8 · TX_DRV · YELLOW; CONSOLE.6 · through 10 kΩ · GPIO18_RX · GREEN; PIN3 · through 10 kΩ · GPIO16_RX · GREEN

Build — exact pin/net/color wiring

Step	Ref	Color	From	To	Part	Note
1	NEW C11-1	NO WIRE	J1/J2 terminals	unwired	NO WIRE	Isolated mapping first: keep all J1/J2 terminals unwired; map each conductor independently and verify unrelated pins open using NO WIRE.

Independent isolated-map evidence — complete before Step 2

CONSOLE terminal map				MOTOR terminal map				Unrelated pins open / operator initials
1	2	3	4	5	6	7	8	
1	2	3	4	5	6	7	8	

Step	Ref	Color	From	To	Part	Note
2	1	BLACK	CONSOLE RJ45 pin 1	GND	Jumper wire	First ground Common only after isolated-map PASS. Connect J1 pin 1 to GND using BLACK; this begins the join pins 1/7 and 2/8 commoning sequence.
3	2	BLACK	MOTOR RJ45 pin 1	GND	Jumper wire	First ground Connect J2 pin 1 to GND using BLACK.
4	3	BLACK	CONSOLE RJ45 pin 7	GND	Jumper wire	Second ground Connect J1 pin 7 to GND using BLACK.
5	4	BLACK	MOTOR RJ45 pin 7	GND	Jumper wire	Second ground Connect J2 pin 7 to GND using BLACK.
6	5	RED	CONSOLE RJ45 pin 2	+8V_RAW	Jumper wire	First +8 V contact Connect J1 pin 2 to +8V_RAW using RED.
7	6	RED	MOTOR RJ45 pin 2	+8V_RAW	Jumper wire	First +8 V contact Connect J2 pin 2 to +8V_RAW using RED.
8	7	RED	CONSOLE RJ45 pin 8	+8V_RAW	Jumper wire	Second +8 V contact Connect J1 pin 8 to +8V_RAW using RED.
9	8	RED	MOTOR RJ45 pin 8	+8V_RAW	Jumper wire	Second +8 V contact Connect J2 pin 8 to +8V_RAW using RED.
10	9	WHITE	CONSOLE RJ45 pin 3	MOTOR RJ45 pin 3	Jumper wire	Direct pass-through Connect CONSOLE.3 endpoint to MOTOR.3 using WHITE.
11	10	WHITE	CONSOLE RJ45 pin 4	MOTOR RJ45 pin 4	Jumper wire	Direct pass-through Connect CONSOLE.4 endpoint to MOTOR.4 using WHITE.
12	11	WHITE	CONSOLE RJ45 pin 5	MOTOR RJ45 pin 5	Jumper wire	Direct safety pass-through Connect CONSOLE.5 endpoint to MOTOR.5 using WHITE.
13	12	YELLOW	CONSOLE RJ45 pin 6	K1 G5V-2 pin 4	Jumper wire	Relay NC contact Connect CONSOLE.6 endpoint to K1 pin 4 using YELLOW.
14	13	YELLOW	MOTOR RJ45 pin 6	K1 G5V-2 pin 6	Jumper wire	Relay COM contact Connect MOTOR.6 endpoint to K1 pin 6 using YELLOW.
15	14	YELLOW	K1 G5V-2 pin 8	R_TX 100 Ω, end 2	Jumper wire	Relay NO / ESP transmit Connect K1 pin 8 to TX_DRV using YELLOW.
16	101	GREEN	CONSOLE RJ45 pin 6	R_CONS_RX 10 kΩ, end 1	Wire	Console RX tap Connect CONSOLE.6 endpoint to K1 pin 4 using YELLOW. Connect CONSOLE.6 through 10 kΩ to GPIO18_RX using GREEN.
17	102	GREEN	R_CONS_RX end 2	DevKit GPIO18	Wire	Console RX Connect CONSOLE.6 through 10 kΩ to GPIO18_RX using GREEN.
18	103	GREEN	RJ45 pin-3 net	R_PIN3_RX 10 kΩ, end 1	Wire	Pin-3 tap Connect PIN3 through 10 kΩ to GPIO16_RX using GREEN.
19	104	GREEN	R_PIN3_RX end 2	DevKit GPIO16	Wire	Pin-3 RX Connect PIN3 through 10 kΩ to GPIO16_RX using GREEN.

Unpowered test

Instruction / kind / points / expected	Evidence
Map every connector conductor independently · mapping · CONSOLE.1–8 to MOTOR.1–8 and K1 pins 4/6/8 · each intended path below 2 Ω; post-common cross-shorts absent except pins 1/7 GND and pins 2/8 +8V_RAW; end-to-end CONSOLE.6 ↔ MOTOR.6 through NC	

Powered test

Instruction / input / output / evidence / limits	Measured
Run bypass-only treadmill sequence · treadmill +8 V through fused harness · current/drop/receive-only UART/thermal · record source state, harness, restored direct paths, endpoints · separately measure treadmill current no more than 500 mA; supply drop no more than 50 mV; ground-return drop no more than 50 mV; 15-minute endpoints no more than 40°C and no more than 10°C over ambient	

Firmware identity: _____ **Observer identity:** _____
Observer manifest: _____ (path or SHA proving relay and TX outputs disabled)

Bypass-only controlled sequence

- 1. All sources off before RJ45 changes. Attach cables only after Clusters 1–10 signed PASS.
- 2. Verify bypass-only source state: USB absent; STANDALONE POWER installed; observer verified; relay off; TX disabled.
- 3. With all sources off before harness changes, remove both independent direct +8V paths and install the reviewed dual-pin fused-DMM harness. Measure current separately at no more than 500 mA.
- 4. Remove power before removal of the harness. Verify zero current, remove harness, then restore both independent +8V paths and continuity-check each.
- 5. Voltage drop is measured only after direct-path restoration. Thermal test only after direct-path restoration; then run the fifteen-minute hold.

Cluster 11 — dedicated bypass and thermal evidence

Evidence field	Measured	Limit/state
Treadmill current (mA)		≤500 mA, fused harness
Supply drop (mV)		≤50 mV after direct paths restored
Ground-return drop (mV)		≤50 mV after direct paths restored

RJ45 pin	CONSOLE temperature (°C)	MOTOR temperature (°C)	Limit after 15 minutes
1 — GND			≤40°C; ≤10°C rise
2 — +8 V			≤40°C; ≤10°C rise
3			≤40°C; ≤10°C rise
4			≤40°C; ≤10°C rise
5			≤40°C; ≤10°C rise
6 — switched data			≤40°C; ≤10°C rise
7 — GND			≤40°C; ≤10°C rise
8 — +8 V			≤40°C; ≤10°C rise

Relevant breadboard power endpoint	Measured	Limit
+8V endpoint temperature (°C): hottest exact endpoint _____		≤40°C; ≤10°C rise
GND endpoint temperature (°C): hottest exact endpoint _____		≤40°C; ≤10°C rise

The current no-control diagnostic cannot pass the relay exercise gate and cannot pass the UART exercise gate. This is observation only. No treadmill transfer and no MCU TX. A future procedure must require qualified functional-firmware identity and matching production safety evidence; bypass PASS grants no transfer permission.

STOP gate: Any map, identity, current, drop, receive-only UART, bypass, thermal, or reset check fails. Power off; do not transfer the relay or transmit.

PASS gate: All eight paths, NC bypass, current, restored-path drop, receive-only observation, and thermal evidence pass. **PASS — continue** only to documentation closeout, never to treadmill control.

Operator: _____

Date: _____

Signed PASS: _____